

Overlap must survive every trimming decision

Budget = retained R1 + retained R2 – biological insert; DADA2 requires at least 12 nt here.

A Selected 220 + 200

Retained read budget = 420 nt

Modal overlap = 167 nt

PASS

Observed insert range: 253 nt mode to 303 nt maximum; minimum observed overlap estimate = 117 nt.

B Overtrimmed 130 + 120

Retained read budget = 250 nt

Modal overlap = -3 nt

FAIL

Observed insert range: 253 nt mode to 303 nt maximum; minimum observed overlap estimate = -53 nt.

A quality plot can suggest a cut; only the overlap equation can approve it.